

Extensions to COST 2100 Channel Model for Extremely Large-Scale MIMO

1st Li Fan

Wireless Network RAN Research
Department
Shanghai Huawei Technologies
CO.,Ltd
Shanghai, China
fanli4@huawei.com

2nd Zhimeng Zhong

Wireless Network RAN Research
Department
Shanghai Huawei Technologies
CO.,Ltd
Shanghai, China
zhongzhimeng@huawei.com

3rd Chao Wang

Wireless Network RAN Research
Department
Shanghai Huawei Technologies
CO.,Ltd
Shanghai, China
wangchao307@huawei.com

4th Qibo Qin

Wireless Network RAN Research
Department
Shanghai Huawei Technologies
CO.,Ltd
Shanghai, China
qinqibo1@huawei.com

5th Wei Han

Wireless Network RAN Research
Department
Shanghai Huawei Technologies
CO.,Ltd
Shanghai, China
wayne.hanwei@huawei.com

6th Tengjiao Wang

Wireless Network RAN Research
Department
Shanghai Huawei Technologies
CO.,Ltd
Shanghai, China
wangtengjiao6@huawei.com

Abstract—Extremely large-scale multiple-input multiple-output (XL-MIMO) system providing more degrees of freedom and beamforming gain, can effectively increase the capacity and coverage of wireless networks, which has become one of the key technologies for 6G. However, the near-field and non-stationary effects will appear as the array aperture expands. Therefore, existing channel models such as COST 2100 are extended to model these propagation effects. Considering the actual deployment of extremely large-scale antennas, this paper further extends the COST 2100 channel, including a) extending BS VR to 3D modeling to support the non-stationary characteristics of large-scale arrays in vertical and horizontal dimensions; b) extending the distribution of MS VR in vertical height to support user equipments (UEs) which are distributed on floors at different heights; c) extending models to support spatially consistent propagation characteristics between multi-TRPs. Finally, we show that our modeling method is reasonable through parameter tuning and simulation, and it can be used to evaluate the system performance of large-scale MIMO with different deployment morphologies.

Keywords—COST 2100 channel model, XL-MIMO, non-stationary effect, near-field propagation

I. INTRODUCTION

Massive multiple-input multiple-output (MIMO), a key technology for increasing area spectral efficiency in cellular systems, has become a reality for the fifth-generation (5G) wireless networks [1]. Looking forward towards the sixth generation (6G), to meet the demands of a fully connected, intelligent digital world, further scaling up the antenna number/size significantly is proposed, known as extremely large-scale MIMO (XL-MIMO) [2]. Compared with traditional MIMO, as the antenna size further increases, XL-MIMO systems behave differently due to spatial non-stationarity and

near-field propagation. In the large-scale regime, non-stationarity occurs because the terminals/mobile stations (MSs) may only see a portion of array at base station (BS) [3]. And MS locates in the near-field region of BS array, in which the commonly used far-field assumption with uniform plane wave (UPW) model may become invalid, instead the near-field radiation with spherical wavefront need to be considered [4]. In order to evaluate the performance of XL-MIMO effectively, the channel model must model the characteristics of spatial non-stationary and spherical wavefront. There are some preliminary efforts on the channel modeling and validation of XL-MIMO channel models [5-7]. Among them, the channel model based on COST 2100 is widely investigated since it naturally supports spherical wave characteristics and even validated based on measurement by research team from Lund. For example, [5] [6] extended a concept of base station-side visibility region (BS-VR) in COST 2100 channel model to model the appearance and disappearance of clusters in XL-MIMO systems and extracted the model parameters for the extensions using a 128 antenna elements uniform linear array (ULA) at 2.6 GHz. In addition to MS-side visibility region (MS-VR), a multipath component visibility region (MPC-VR) is proposed in [6] to model the birth-death processes of individual MPCs. However, with these extensions, COST 2100 channel model still has the following limitations: a) MS-VR is modeled as a circle that is evenly distributed on the horizontal plane and cannot support simulation of MSs distributed on different floors, which is a typical simulation scenario. b) BS-VRs are valid only for extremely large-scale ULA. But in practice, the array is mostly a uniformly planar array (UPA) with elements distributed both horizontally and vertically and can be deployed along the wall to form an extremely large-scale planar array. c) The model

cannot support multiple arrays deployed in the same location but with different boresight.

This paper further extends the COST 2100 channel model to break the above limitations. In this paper, we first briefly introduce the general framework of COST 2100 channel model in Section II, then describe the modeling extensions in Section III, and give some simulation results to validate the extensions in Section IV, and finally make a conclusion in Section V.

II. OVERVIEW OF COST 2100 CHANNEL MODEL

Firstly, we briefly review some concepts and terminologies of the COST 2100 model. And more details can be found in [8] [9]. The COST 2100 model falls into the category of geometry-based stochastic channel models (GSCM), and more specifically it belongs to the cluster-level channel model, differs from the system-level channel model, such as 3GPP channel model in [10]. In the cluster-level COST 2100 modeling process [11], the modeling process is specified once for the entire environment by: a) defining a large quantity of clusters with consistent stochastic parameters throughout the simulation environment based on the BS location (yet, not all clusters are visible at any time instant), b) defining the MS location and determining the scattering from the so-called visible clusters at each channel instance, c) synthesizing the channel parameters based on the cluster scattering. Fig. 1 depicts the scattering mechanisms from the BS to the MS. The far clusters are classified into two types: single cluster and twin cluster, to model the single bounce cluster and multi-bounce clusters, respectively. Another distinguishing feature is the visibility regions, each associated to a cluster. The visibility regions are used to model non-stationarity in a spatiotemporally consistent way. In Fig. 1, two types of VRs are given. One is MS-VR and the other is the BR-VR. In the XL-MIMO scenario, a double directional impulse response exists between the BS array antenna element and the MS only when the antenna element and the MS are located in the same cluster corresponding to the BS VR and MS VR.

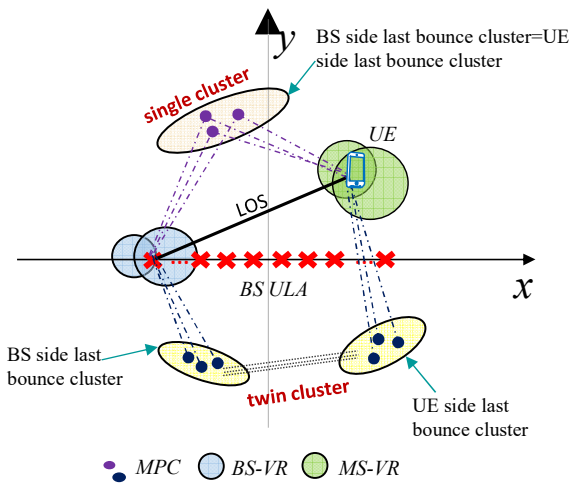


Fig. 1. The general description of COST 2100 channel model.

III. PROPOSED CHANNEL MODEL EXTENSIONS

In this section we introduce our extensions of the COST 2100 channel model.

A. Extension to 3-Dimensional(3D) BS VR

BS VR is modelled in 2 dimension (2D) space as a circle [6]. Specifically, [6] models the center of BS VR long ULA, with exponentially distributed lifetimes, and that the number of BS-VRs is Poisson distributed with intensity proportional to the sum of the array length and the mean lifetime. In this modelling approach, BS VR can be considered as a line segment distributed along the ULA as shown in Fig. 2(a). We model the BS VRs as an ellipsoid instead of a line as shown in Fig. 2(b), so that the array antenna element can be arranged in an arbitrary three dimension space. Firstly, we should determine the distribution of BS VRs' center and length of each axis which models the lifetime of the BS VR. The BS VRs' center is uniformly distributed along the antenna array's surface, and the lifetimes of BS VRs remains modeled exponentially as in [6]. The remaining parameters of the BS VR ellipsoid to be determined are the expectations of the length of each axis.

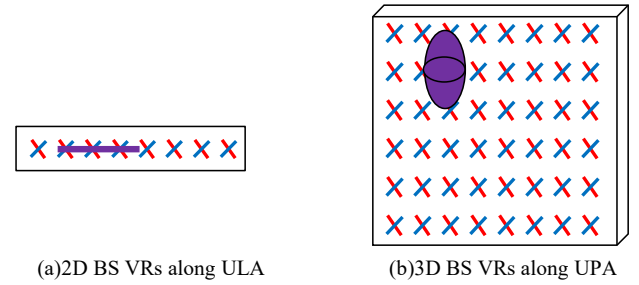


Fig. 2. The modeling of 3D BS VRs.

Let a, b and c be the principal semiaxes of BS VR ellipsoid shown in Fig. 3. First assume that $a = b$, because both a and b describe the visibility of clusters on the horizontal plane, it can be considered that the horizontal propagation environment is statistically isotropic. But in general, the propagation environment in the vertical dimension is different from that in the horizontal dimension, which leads to the modeling of $a = b \neq c$. Then, we mainly determine the radius of ellipse e on the YoZ plane of the local coordinate system, as shown in Fig. 3. The BS VRs distributed along ULA can be considered as a special case of 3D BS VRs. Generally, the ULA does not always go through the center of the 3D BS VR. We propose two methods for finding the relation between the value of a, c and the lifetime distance of BS VR along ULA as follows.

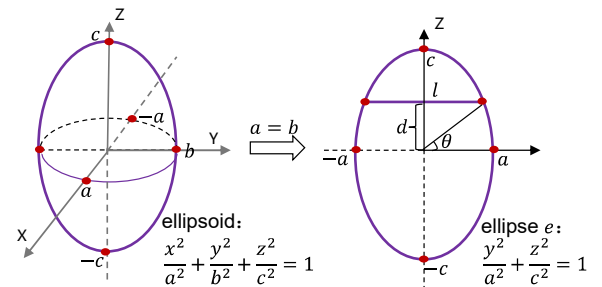


Fig. 3. Modeling the BS VR as an ellipsoid.

1) Modeling Method 1

Assume that the ULA is a segment of a line that is observed horizontally uniformly on the UPA. (Endpoints are evenly distributed on the ellipse, that is, endpoints of the line segment parallel to the Y axis in Fig. 3 are evenly distributed on the ellipse e). Given this geometry, the length of an intersection between ULA and the ellipse is

$$l = \begin{cases} 2|y| & y \text{ on ellipse } e \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

The average of l is

$$L = E[l] = \int_{-a}^a 2|y| f_e(y) dy \quad (2)$$

where $E[\cdot]$ denotes statistical expectation and $f_e(y)$ is the probability density function corresponding to the uniform distribution of y on the ellipse e . Under the condition that L is known, a in (2) can be solved by using Monte Carlo simulation. For example, for a typical value 3.2 meter in [6], a is equal to 2.52 meter. In particular, when the ellipse degenerates to a circle, (2) can be simplifies as:

$$L = E[l] = \frac{2}{\pi} \int_0^{\pi/2} 2a \cos(\theta) d\theta = \frac{4a}{\pi} \quad (3)$$

2) Modeling Method 2

In method 2, it's assumed that the ULA traverses the ellipse visibility regions at a random (uniformly distributed) distance d from the respective centers of the cluster visibility regions. Given this geometry, the length of an intersection between the ULA and the ellipse is

$$l = \begin{cases} 2a \sqrt{1 - \frac{d^2}{c^2}} & 0 \leq d \leq c \\ 0 & \text{otherwise} \end{cases} \quad (4)$$

The mean value of l is

$$L = E[l] = \int_0^c 2a \sqrt{1 - \frac{z^2}{c^2}} f_d(z) dz \quad (5)$$

where $f_d(z)$ is the probability density function for d . By solving the integral in (5) for a uniform distributed d , $0 \leq d \leq c$, we obtain

$$L = \frac{\pi a}{2} \quad (6)$$

From (6), when $L = 3.2$ meter, $a = 2.04$ meter.

It can be found from the above that the value of a obtained in Model 1 is greater than the value of Model 2. The smaller the value of a , the more significant the non-stationary phenomenon is. Under the same number of BS VRs, the density of BS VRs on the array decreases after the BS VRs are expanded from the distribution on the ULA to the distribution on the UPA. In order to maintain the same BS VR density, the number of BS VRs

needs to be increased. In the current model, the number of BS VRs is the same as the number of clusters. Increasing the number of clusters will lead to lower simulation efficiency. Therefore, it is recommended to use Model 1 for modeling by default.

Since the method for obtaining c is similar to a , the details are omitted here. Currently, there is no XL-MIMO measurement data in the vertical dimension. Therefore, only one empirical formula is provided as

$$c = \gamma \cdot a \quad (7)$$

In a typical urban macro (UMA) propagation environment, the spread of channel parameters in vertical dimension is less than that in the horizontal dimension. Therefore, the non-stationary feature of the vertical dimension is weaker than that of the horizontal dimension, meaning $\gamma > 1$, e.g. $\gamma = 2$. If there is an XL-MIMO measurement data of the vertical dimension, the specific value of γ can be modified.

In conclusion, the BS VR is modeled as an ellipsoid whose center is uniformly distributed on the surface of the antenna array. The length, width, and height of the ellipsoid are exponential distributions whose mean values are $2a$, $2b$, and $2c$ respectively.

B. Extension to UE Distributed on Different Floors

In COST 2100 channel model, MS VRs are identically sized circular regions on the azimuth plane, which only support the horizontal distribution and movement of the MS. In typical simulation scenarios such as in [10], 80% of users are randomly distributed in buildings with floors ranging from 4 to 8. For XL-MIMO, MSs should not be limited to only horizontal distribution, but needs to support vertical distribution in different floors. Therefore, it is necessary to extend the modeling of MS VS in COST 2100.

A relatively direct modeling method is to expand the MS VR distributed in the horizontal plane into a sphere or ellipsoid distributed in a three-dimensional space, similar to 3D BS VR. With such modeling, UEs between different floors may observe the same cluster, meaning that the environment has spatial consistency between different heights. In the current popular 3GPP channel model [10], the channels between different floors are independent of each other and have no spatial consistency. And there is no measured data to show the proportion of common clusters between floors. Therefore, this modeling method has the disadvantage that the parameters are difficult to determine. Therefore, for the case where MSs are distributed at different floor heights, a simpler modeling method can be adopted. That is, each floor independently distributes its corresponding MS VR. When MS VRs are in one-to-one correspondence with remote clusters, assuming that the highest floor is N , the total number of MS VRs is N times higher than that of pure outdoor distribution. From the perspective of the geometric distribution of MS VR, it is equivalent to modeling MS VR as circles distributed on the planes corresponding to different building heights.

C. Extension to Co-sited Multi-Transmission and Reception Point (TRP)

Accurate modeling of radio wave propagation in multilink scenarios is crucial for system-level simulations of MIMO systems. An extension of the COST 2100 channel model to a multilink level is proposed in [12]. Focus on the BS side, the BS-common cluster (BSCC) is introduced to model the correlation between BSs. But the ratio of BSCC is difficult to get, and only one indoor measurement result is provided in [13]. In 5G new radio system, one site can deploy several TRPs. These co-sited TRPs can be assumed to share the same propagation environment, but with different boresight. Without introducing common clusters between TRPs, we propose another modeling method. For the example in Fig. 4, three TRPs are located on one site with boresight of 90° , -150° , and -30° respectively. Specific modeling steps include:

1) *Average number of visible far clusters*: Compared with single TRP, multi-TRPs increase the antenna aperture and can distinguish more clusters in the environment. Therefore, the number of clusters corresponding to M -TRPs can be adjusted to M times that of one TRP. In COST 2100 channel model, N_c denotes the average number of visible far clusters. In [6], for single ULA, N_c in Line Of Sight (LOS) scenario is fixed as 10, and in Non-LOS (NLOS) is

$$N_c = 2.9 \times (L_{BS} + L) \quad (8)$$

where L_{BS} is the length of ULA, and L is the mean value of BS VR lifetime distance. When extended to UPA, N_c needs to be modified. Aiming at keeping the same number of active clusters observed by a single antenna element before and after model expansion, N_c for NLOS is adjusted by Monte Carlo simulation as

$$N_c = 2.9 \times (L_{BS,H} + L) \times \max\{1, L_{BS,V}/8\} \quad (9)$$

where $L_{BS,H}$ and $L_{BS,V}$ denote the lengths of the UPA in the horizontal and vertical dimensions, respectively. For the LOS case, N_c is set to 0.6 of the value corresponding to NLOS by referring to the UMA scenario in [10].

2) *Cluster distribution*: Calculate the centroid of multiple TRPs and use the connection between the centroid and the MS VR center as a reference to generate the location of the far clusters in the environment. As depicted in Fig. 4, the red circle denotes the centroid of multi-TRPs, and the dash line connected TRPs' centroid and MS VR's center is used to place clusters. The detailed modeling of clusters are kept the same as in basic COST 2100 channel model [13], which is omitted here.

3) *3D BS VR distribution*: BS VRs are modeled as ellipsoid as described above. Their centers are uniformly distributed on the surface composed of multi-TRPs and the length, width and height satisfy the exponential distribution with mean value of $2a$, $2b$ and $2c$.

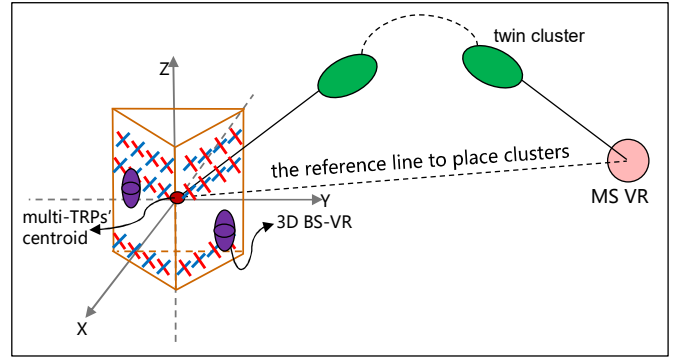


Fig. 4. Modeling of 3-TRPs in one site.

IV. SIMULATION RESULTS AND ANALYSIS

Based on the open source COST 2100 channel model [15], the above extensions are added. In this chapter, the parameter table before and after the model expansion is given first, and then the statistics corresponding to delay spread and angular spread are listed.

A. Simulation Settings

We adjusted the parameters of the COST 2100 channel according to some measured data, some parameters are shown in Table I, and the rest of the parameters mainly refer to the configuration of the open source code [15] and the paper [6]. In the simulation, the center frequency is 3.5 GHz, the bandwidth is 20 MHz. A MIMO UPA with 4 rows and 8 columns patches was used at BS. Each antenna patch is separated by half wavelength at 3.5 GHz, and has a pair of $\pm 45^\circ$ polarized co-located antenna elements.

TABLE I. PARAMETRIZATION OF THE COST 2100 MODEL EXTENSIONS

Model Parameters	Before Extension		After Extension	
	LOS	NLOS	LOS	NLOS
Average number of visible far clusters	10	$2.9 \times (L_{BS} + L)$	$1.74 \times (L_{BS,H} + L) \times \max\{1, L_{BS,V}/8\}$	$2.9 \times (L_{BS,H} + L) \times \max\{1, L_{BS,V}/8\}$
Cluster power decay factor [dB/ μ s]	30	3	30	3
Cluster shadowing	3	4	3	4
Cluster delay spread[mean, std]	0.01, 3.2	0.14, 2.85	0.01, 3.2	0.14, 2.85
Cluster angular spread in azimuth at BS[mean, std]	5,2.93	5,2.4	5,2.93	5,2.4
Cluster angular spread in elevation at BS[mean, std]	0,0	1,0	0,0	1,0
Cluster angular spread in azimuth at MS[mean, std]	5,2.68	10,2.03	5,2.68	10,2.03
Cluster angular spread in elevation at MS[mean, std]	0,0	1.578,0	0,0	1.578,0

B. Simulation Results

The delay and angular spreads are important statistical characteristics of the channel model. These spread values before and after our expansion are listed in Table II. It can be found that after expansion, the delay and azimuth angular spread under LOS scenario decreases, and has a larger gap with NLOS scenario, which is more consistent with the law of channel propagation. The elevation angular spread does not change much, which is probably because the size of the configured antenna array is not large in the vertical size, and the non-stationary effect of the vertical plane is not obvious. In summary, after the model is expanded, the difference is small under the typical configuration, but it can additionally support more simulation requirements.

TABLE II. PARAMETRIZATION OF THE COST 2100 MODEL EXTENSION FOR ELAA AT 3.5 GHz

Parameters' Spread	Before Extension				After Extension			
	LOS		NLOS		LOS		NLOS	
DS(ns)	205	119	218	94	140	62	224	86
ASA(degree)	35	29	61	22	25	11	64	23
ASD(degree)	33	29	27	10	15	7	28	10
ESA(degree)	14	9	13	6	11	5	14	5
ESD(degree)	6	5	9	3	5	2	9	3

V. CONCLUSION

This paper extends the COST 2100 channel model, including a) extending BS VR to 3D modeling to support the non-stationary characteristics of large scale arrays in vertical and horizontal dimensions; b) extending the distribution of MS VR in vertical height to support UEs are distributed on floors at different heights; c) extending models to support spatially consistent propagation characteristics between multi-TRPs. The channel model parameters are properly adjusted for extensions, and finally the statistical spread of the channel model before and after the expansion are compared. It can be seen that after the model is extended, the relative relationship of the spread between LOS and NLOS case is more in line with the channel propagation rule.

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